

Portfolio - Astha Yadav

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OBJECTIVE

I am a Physics Major whose focus is in Quantum Communication, intertwining Quantum Physics with cryptography and network security.

TECHNICAL SKILLS

- **Quantum Security & Simulation:** Qiskit, QNetSim, QuREBB, Quantum Key Distribution (QKD), Decoy-State Protocols
- **Programming & Tools:** Python, C, MATLAB, SQL, HTML/CSS, Git/GitHub, $\LaTeX$
- **Data Science & Numerical Methods:** NumPy, Pandas, Scikit-learn, Matplotlib, Euler/RK4 Integration, Finite Difference Methods
- **Core Cybersecurity:** Network Security Infrastructure, Cyber Forensics, Anomaly Detection Basics

WORK EXPERIENCE

- **Co-Author & Research Assistant — Quantum Chemistry Simulation** March 2026 - May 2026
Collaborative Research Project Fujitsu Quantum Simulator Challenge 2026
 - Worked in a multidisciplinary team to execute 16-qubit simulations on IQM's Sirius 24-qubit superconducting processor, mapping the 2D potential energy surface (32×32 grid) of H_2O
 - Co-authored a research article demonstrating the integration of Density Matrix Embedding Theory (DMET) to scale fragment-based molecular simulations toward industrial limits while maintaining chemical accuracy
 - Formulated a comprehensive literature review on Sample-based Quantum Diagonalization (SQD) frameworks, establishing the theoretical baseline for non-local molecular electronic structure simulations
 - We evaluated the ansatz expressivity and gate-depth trade-offs between hardware-efficient LUCJ and chemically motivated LCNot-UCCSD.

ACADEMIC RESEARCH & PROJECTS

- **Bachelor's Thesis - Quantum Cryptography (Applications, Security of QKD)** August 2025 - April 2026
Advisor: Prof. Bipin Singh Koranga Kirori Mal College
 - QKD protocols; deployment cases; simulating implementation through qiskit & analysing QBER
 - Mastery of concepts including density matrices, quantum error correction principles, and key quantum algorithms.
 - Quantum Network simulations using QNetSim; QuREBB library
 - simulating fiber attenuation and detector noise; modeling vulnerabilities of multi-photon pulses under PNS attacks
 - studying how decoy-state protocols and machine learning could help in error mitigation (latter for CVQKD, focusing on elimination environmental drift)
 - Skills: Quantum Information Theory, Python, Qiskit

AWARDS & COMPETITIVE ACHIEVEMENTS

- **Winner of BB84 Challenge | IBM Qiskit Fall Fest '25 (PES University)** Nov 2025
- **Participant | MIT iQuHack 2026 (IonQ Entanglement Challenge)** Feb 2026

TECHNICAL CERTIFICATIONS & DIPLOMAS

- **QWorld Diplomas:** [QSilver](#) (March 2026) | [QBronze](#) (July 2024)
- **FDP MeitY:** [Quantum Computing & Post-Quantum Cryptography](#) Feb 2025
- **IBM Quantum:** [Qiskit Global Summer School 2025 Badge](#) Aug 2025
- **WISER Summer Program:** [Quantum Algorithms](#) | [Non-Linear Problems](#) July 2025
- **Short Courses:** [IQM Quantum School](#) | TryHackMe Cybersec Challenge Dec 2025

EDUCATION

- **B.Sc. (Hons. with Research) in Physics, with Minor in Computer Science** November 2022 - June 2026
Kirori Mal College, University of Delhi Delhi, India
 - Percentage: 72%
- **Senior-Secondary/12th Grade** April 2022
CBSE Board Delhi, India
 - Percentage: 77.6%
- **Secondary/10th Grade** April 2020
CBSE Board Delhi, India
 - Percentage: 96%

ADDITIONAL ACHIEVEMENTS

Languages: English (College Method of Instruction), French (A1 Certificate), Hindi & Awadhi (Native)
Professional Memberships: American Physical Society, QIndia
Leadership Roles: Group Leader in Communications Systems Semester Project (Sem-5), Student Council of FilmSOC

B.SC. (HONS) PHYSICS COURSEWORK (MAJOR)

Theoretical Physics: Classical Mechanics (Lagrangian & Hamiltonian), Quantum Mechanics I & II (Hilbert Spaces, Angular Momentum, Density Matrices), Advanced Quantum Mechanics (Time-Dependent Perturbation, Fermi's Golden Rule, Scattering Theory, Born Approximation, Variational Methods), Statistical Mechanics & Advanced Statistical Mechanics (Grand Canonical Ensembles, Ising Model, Quantum Density Matrices), Electromagnetic Theory (Maxwell's Eqns, Wave Guides), Special Relativity, and Modern Physics.
Mathematical & Computational: Mathematical Physics I-III (Complex Analysis, PDEs, Fourier Transforms), Linear Algebra, Numerical Methods, Bayesian Statistics, Python libraries (Pandas, Seaborn, NumPy, Scikit-learn, Matplotlib)
Applied & Experimental: Solid State Physics (Brillouin Zones, Hall Effect), Atomic & Molecular Physics, Analog & Digital Electronics (BJTs, Op-Amps, Logic Circuits), Communication Systems, Waves and Optics, and Thermal Physics.
Ancillary: DBMS, Operating Systems, Web Development, and Research Methodology (Ethics & IPR).

COMPUTER SCIENCE COURSEWORK (MINOR)

Core Systems & Architecture: Operating Systems (Process Scheduling, Paging, Shell Scripting), Database Management Systems (ER Modeling, SQL, 1NF/2NF/3NF Normalization), and Web Programming (HTML/CSS, JavaScript, jQuery, JSON).
Data Structures & Algorithms: Data Structures Using Python (Linked Lists, Stacks, Queues, Binary Search Trees, Binary Heaps), Growth of Functions, and Recurrence Relations.
Intelligence & Language Modeling: Machine Learning (PCA, Regression, SVM, Back-propagation Neural Networks, Clustering) and Natural Language Processing (TF-IDF, Word2vec, RNNs/LSTMs, Transformers, BERT, GPT).
Security & Forensics: Information Security (Symmetric/Public-Key Cryptography, PKI, RBAC/ABAC, SQL Injection, Firewalls/DDoS Defenses, Cloud/IoT Security) and Cyber Forensics (Evidence Recovery, Hive File Analysis, Autopsy/EnCase/FTK, IT Laws).